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According to the passage, each of the following is true of endangered amphibian species EXCEPT:

A. They are among the rarest kinds of amphibians.

B. They generally have populations that are small in size.

C. They are in constant danger of extinction.

D. Those with decreasing populations are the most likely candidates for immediate extinction.

E. They are in danger of extinction due to events that sometimes have nothing to do with human activities.

•

Which of the following most accurately describes the organization of the passage?

A. A question is raised, a distinction regarding it is made, and the question is answered.

B. An interpretation is presented, its soundness is examined, and a warning is given.

C. A situation is described, its consequences are analyzed, and a prediction is made.

D. Two interpretations of a phenomenon are described, and one of them is rejected as invalid.

E. Two methods for analyzing a phenomenon are compared, and further study of the phenomenon is recommended.

•

Which of the following best describes the function of the sentence in lines 35–38?

A. To give an example of a particular kind of study

B. To cast doubt on an assertion made in the previous sentence

C. To raise an objection to a view presented in the first paragraph

D. To provide support for a view presented in the first paragraph

E. To introduce an idea that will be countered in the following paragraph

Line While the most abundant and dominant species within a particular ecosystem is often crucial in perpetuating the ecosystem, a "keystone" species, here defined as one whose effects are much larger (5) than would be predicted from its abundance, can also play a vital role. But because complex species interactions may be involved, identifying a keystone species by removing the species and observing changes in the ecosystem is problematic. It might (10) seem that certain traits would clearly define a species as a keystone species; for example, *Pisaster ochraceus* is often a keystone predator because it consumes and suppresses mussel populations, which in the absence of this starfish (15) can be a dominant species. But such predation on a dominant or potentially dominant species occurs in systems that do as well as in systems that do not have species that play keystone roles. Moreover, whereas *P. ochraceus* occupies an unambiguous (20) keystone role on wave-exposed rocky headlands, in more wave-sheltered habitats the impact of *P. ochraceus* predation is weak or nonexistent, and at certain sites sand burial is responsible for eliminating mussels. Keystone status appears to (25) depend on context, whether of particular geography or of such factors as community diversity (for example, a reduction in species diversity may thrust more of the remaining species

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diversity may thrust more of the remaining species into keystone roles) and length of species
(30) interaction (since newly arrived species in particular may dramatically affect ecosystems).

**Questions 587–590 refer to the passage.**

587. The passage mentions which of the following as a factor that affects the role of *P. ochraceus* as a keystone species within different habitats?

A. The degree to which the habitat is sheltered from waves

B. The degree to which other animals within a habitat prey on mussels

C. The fact that mussel populations are often not dominant within some habitats occupied by *P. ochraceus*

D. The size of the *P. ochraceus* population within the habitat

E. The fact that there is great species diversity within some habitats occupied by *P. ochraceus*

588. Which of the following hypothetical experiments most clearly exemplifies the method of identifying species' roles that the author considers problematic?

A. A population of seals in an Arctic habitat is counted in order to determine whether it is the dominant species in that ecosystem.

B. A species of fish that is a keystone species in one marine ecosystem is introduced into another marine ecosystem to see whether the species will come to occupy a keystone role.

C. In order to determine whether a species of monkey is a keystone species within a particular ecosystem, the monkeys are removed from that ecosystem and the ecosystem is then studied.

D. Different mountain ecosystems are compared to determine how geography affects a particular species' ability to dominate its ecosystem.

E. In a grassland experiencing a changing climate, patterns of species extinction are traced in order to evaluate the effect of climate changes on keystone species in that grassland.

589. Which of the following, if true, would most clearly support the argument about keystone status advanced in the last sentence of the passage (lines 24–31)?

A. A species of bat is primarily responsible for keeping insect populations within an ecosystem low, and the size of the insect population in turn affects bird species within that ecosystem.

B. A species of iguana occupies a keystone role on certain tropical islands but does not play that role on adjacent tropical islands that are inhabited by a greater number of animal species.

C. Close observation of a savannah ecosystem reveals that more species occupy keystone roles within that ecosystem than biologists had previously believed.

D. As a keystone species of bee becomes more abundant, it has a larger effect on the ecosystem it inhabits.

E. A species of moth that occupies a keystone role in a prairie habitat develops coloration patterns that camouflage it from potential predators.

590. The passage suggests which of the following about the identification of a species as a keystone species?

A. Such an identification depends primarily on the species' relationship to the dominant species.

B. Such an identification can best be made by removing the species from a particular ecosystem and observing changes that occur in the ecosystem.

C. Such an identification is likely to be less reliable as an ecosystem becomes less diverse.

D. Such an identification seems to depend on various factors within the ecosystem.

E. Such an identification can best be made by observing predation behavior.

Line Conodonts, the spiky phosphatic remains (bones and teeth composed of calcium phosphate) of tiny marine animals that probably appeared about 520 million years ago, were once among the most
(5) controversial of fossils. Both the nature of the organism to which the remains belonged and the

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organism to which the remains belonged and the function of the remains were unknown. However, since the 1981 discovery of fossils preserving not just the phosphatic elements but also other remains of the tiny soft-bodied animals (also called conodonts) that bore them, scientists' reconstructions of the animals' anatomy have had important implications for hypotheses concerning the development of the vertebrate skeleton.

The vertebrate skeleton had traditionally been regarded as a defensive development, champions of this view postulating that it was only with the much later evolution of jaws that vertebrates became predators. The first vertebrates, which were soft-bodied, would have been easy prey for numerous invertebrate carnivores, especially if these early vertebrates were sedentary suspension feeders. Thus, traditionalists argued, these animals developed coverings of bony scales or plates, and teeth were secondary features, adapted from the protective bony scales. Indeed, external skeletons of this type are common among the well-known fossils of ostracoderms, jawless vertebrates that existed from approximately 500 to 400 million years ago. However, other paleontologists argued that many of the definitive characteristics of vertebrates, such as paired eyes and muscular and skeletal adaptations for active life, would not have evolved unless the first vertebrates were predatory. Teeth were more primitive than external armor according to this view, and the earliest vertebrates were predators.

The stiffening notochord along the back of the body, V-shaped muscle blocks along the sides, and posterior tail fins help to identify conodonts as among the most primitive of vertebrates. The lack of any mineralized structures apart from the elements in the mouth indicates that conodonts were more primitive than the armored jawless fishes such as the ostracoderms. It now appears that the hard parts that first evolved in the mouth of an animal improved its efficiency as a predator, and that aggression rather than protection was the driving force behind the origin of the vertebrate skeleton.

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**Questions 591–593 refer to the passage.**

591. According to the passage, the anatomical evidence provided by the preserved soft bodies of conodonts led scientists to conclude that

A. conodonts had actually been invertebrate carnivores

B. conodonts' teeth were adapted from protective bony scales

C. conodonts were primitive vertebrate suspension feeders

D. primitive vertebrates with teeth appeared earlier than armored vertebrates

E. scientists' original observations concerning the phosphatic remains of conodonts were essentially correct

592. The second paragraph in the passage serves primarily to

A. outline the significance of the 1981 discovery of conodont remains to the debate concerning the development of the vertebrate skeleton

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B. contrast the traditional view of the development of the vertebrate skeleton with a view derived from the 1981 discovery of conodont remains

C. contrast the characteristics of the ostracoderms with the characteristics of earlier soft-bodied vertebrates

D. explain the importance of the development of teeth among the earliest vertebrate predators

E. present the two sides of the debate concerning the development of the vertebrate skeleton

593. It can be inferred that on the basis of the 1981 discovery of conodont remains, paleontologists could draw which of the following conclusions?

A. The earliest vertebrates were sedentary suspension feeders.

B. Ostracoderms were not the earliest vertebrates.

C. Defensive armor preceded jaws among vertebrates.

D. Paired eyes and adaptations for activity are definitive characteristics of vertebrates.

E. Conodonts were unlikely to have been predators.

Line Because the framers of the United States Constitution (written in 1787) believed that protecting property rights relating to inventions would encourage the new nation's economic growth, they gave (5) Congress—the national legislature—a constitutional mandate to grant patents for inventions. The resulting patent system has served as a model for those in other nations. Recently, however, scholars have questioned whether the American system helped (10) achieve the framers' goals. These scholars have contended that from 1794 to roughly 1830, American inventors were unable to enforce property rights because judges were "antipatent" and routinely invalidated patents for arbitrary reasons. This (15) argument is based partly on examination of court decisions in cases where patent holders ("patentees") brought suit alleging infringement of their patent rights. In the 1820s, for instance, 75 percent of verdicts were decided against the patentee. (20) The proportion of verdicts for the patentee began to increase in the 1830s, suggesting to these scholars that judicial attitudes toward patent rights began shifting then.

Not all patent disputes in the early nineteenth (25) century were litigated, however, and litigated cases were not drawn randomly from the population of disputes. Therefore, the rate of verdicts in favor of patentees cannot be used by itself to gauge changes in judicial attitudes (30) or enforceability of patent rights. If early judicial decisions were prejudiced against patentees, one might expect that subsequent courts—allegedly more supportive of patent rights—would reject the former legal precedents. But pre-1830 (35) cases have been cited as frequently as later decisions, and they continue to be cited today, suggesting that the early decisions, many of which clearly declared that patent rights were a just recompense for inventive ingenuity, (40) provided a lasting foundation for patent law. The proportion of judicial decisions in favor of patentees began to increase during the 1830s

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patentees began to increase during the 1830s because of a change in the underlying population of cases brought to trial. This change was partly (45) due to an 1836 revision to the patent system: an examination procedure, still in use today, was instituted in which each application is scrutinized for its adherence to patent law. Previously, patents were automatically granted upon payment (50) of a $30 fee.

**Questions 594–598 refer to the passage.**

594. The passage implies that which of the following was a reason that the proportion of verdicts in favor of patentees began to increase in the 1830s?

A. Patent applications approved after 1836 were more likely to adhere closely to patent law.

B. Patent laws enacted during the 1830s better defined patent rights.

C. Judges became less prejudiced against patentees during the 1830s.

D. After 1836, litigated cases became less representative of the population of patent disputes.

E. The proportion of patent disputes brought to trial began to increase after 1836.

595. The passage implies that the scholars mentioned in line 8 would agree with which of the following criticisms of the American patent system before 1830?

A. Its definition of property rights relating to inventions was too vague to be useful.

B. Its criteria for the granting of patents were not clear.

C. It made it excessively difficult for inventors to receive patents.

D. It led to excessive numbers of patent-infringement suits.

E. It failed to encourage national economic growth.

596. It can be inferred from the passage that the frequency with which pre-1830 cases have been cited in court decisions is an indication that

A. judicial support for patent rights was strongest in the period before 1830

B. judicial support for patent rights did not increase after 1830

C. courts have returned to judicial standards that prevailed before 1830

D. verdicts favoring patentees in patent-infringement suits did not increase after 1830

E. judicial bias against patentees persisted after 1830

597. It can be inferred from the passage that the author and the scholars referred to in line 21 disagree about which of the following aspects of the patents defended in patent-infringement suits before 1830?

A. Whether the patents were granted for inventions that were genuinely useful

B. Whether the patents were actually relevant to the growth of the United States economy

C. Whether the patents were particularly likely to be annulled by judges

D. Whether the patents were routinely invalidated for reasons that were arbitrary

E. Whether the patents were vindicated at a significantly lower rate than patents in later suits

598. The author of the passage cites which of the following as evidence challenging the argument referred to in lines 14–15?

A. The proportion of cases that were decided against patentees in the 1820s

B. The total number of patent disputes that were litigated from 1794 to 1830

C. The fact that later courts drew upon the legal precedents set in pre-1830 patent cases

D. The fact that the proportion of judicial decisions in favor of patentees began to increase during the 1830s

E. The constitutional rationale for the 1836 revision of the patent system

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E. The constitutional rationale for the 1886 revision of the patent system

Line Jacob Burckhardt's view that Renaissance European women "stood on a footing of perfect equality" with Renaissance men has been repeatedly cited by feminist scholars as a prelude to their (5) presentation of rich historical evidence of women's inequality. In striking contrast to Burckhardt, Joan Kelly in her famous 1977 essay, "Did Women Have a Renaissance?" argued that the Renaissance was a period of economic and social decline for women (10) relative both to Renaissance men and to medieval women. Recently, however, a significant trend among feminist scholars has entailed a rejection of both Kelly's dark vision of the Renaissance and Burckhardt's rosy one. Many recent works by these (15) scholars stress the ways in which differences among Renaissance women—especially in terms of social status and religion—work to complicate the kinds of generalizations both Burckhardt and Kelly made on the basis of their observations about (20) upper-class Italian women.

The trend is also evident, however, in works focusing on those middle- and upper-class European women whose ability to write gives them disproportionate representation in the historical (25) record. Such women were, simply by virtue of their literacy, members of a tiny minority of the population, so it is risky to take their descriptions of their experiences as typical of "female experience" in any general sense. Tina Krontiris, for example, in (30) her fascinating study of six Renaissance women writers, does tend at times to conflate "women" and "women writers," assuming that women's gender, irrespective of other social differences, including literacy, allows us to view women as a homogeneous (35) social group and make that group an object of analysis. Nonetheless, Krontiris makes a significant contribution to the field and is representative of those authors who offer what might be called a cautiously optimistic assessment of Renaissance (40) women's achievements, although she also stresses the social obstacles Renaissance women faced when they sought to raise their "oppositional voices." Krontiris is concerned to show women intentionally negotiating some power for themselves (45) (at least in the realm of public discourse) against potentially constraining ideologies, but in her sober and thoughtful concluding remarks, she suggests that such verbal opposition to cultural stereotypes was highly circumscribed; women seldom attacked (50) the basic assumptions in the ideologies that oppressed them.

**Questions 599–605 refer to the passage.**

599. The author of the passage discusses Krontiris primarily to provide an example of a writer who

A. is highly critical of the writings of certain Renaissance women

B. supports Kelly's view of women's status during the Renaissance

C. has misinterpreted the works of certain Renaissance women

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C. has misinterpreted the works of certain Renaissance women

D. has rejected the views of both Burckhardt and Kelly

E. has studied Renaissance women in a wide variety of social and religious contexts

600. According to the passage, Krontiris's work differs from that of the scholars mentioned in line 12 in which of the following ways?

A. Krontiris's work stresses the achievements of Renaissance women rather than the obstacles to their success.

B. Krontiris's work is based on a reinterpretation of the work of earlier scholars.

C. Krontiris's views are at odds with those of both Kelly and Burkhardt.

D. Krontiris's work focuses on the place of women in Renaissance society.

E. Krontiris's views are based exclusively on the study of a privileged group of women.

601. According to the passage, feminist scholars cite Burckhardt's view of Renaissance women primarily for which of the following reasons?

A. Burckhardt's view forms the basis for most arguments refuting Kelly's point of view.

B. Burckhardt's view has been discredited by Kelly.

C. Burckhardt's view is one that many feminist scholars wish to refute.

D. Burckhardt's work provides rich historical evidence of inequality between Renaissance women and men.

E. Burckhardt's work includes historical research supporting the arguments of the feminist scholars.

602. It can be inferred that both Burckhardt and Kelly have been criticized by the scholars mentioned in line 12 for which of the following?

A. Assuming that women writers of the Renaissance are representative of Renaissance women in general

B. Drawing conclusions that are based on the study of an atypical group of women

C. Failing to describe clearly the relationship between social status and literacy among Renaissance women

D. Failing to acknowledge the role played by Renaissance women in opposing cultural stereotypes

E. Failing to acknowledge the ways in which social status affected the creative activities of Renaissance women

603. The author of the passage suggests that Krontiris incorrectly assumes that

A. social differences among Renaissance women are less important than the fact that they were women

B. literacy among Renaissance women was more prevalent than most scholars today acknowledge

C. during the Renaissance, women were able to successfully oppose cultural stereotypes relating to gender

D. Renaissance women did not face many difficult social obstacles relating to their gender

E. in order to attain power, Renaissance women attacked basic assumptions in the ideologies that oppressed them

604. The last sentence in the passage serves primarily to

A. suggest that Krontiris's work is not representative of recent trends among feminist scholars

B. undermine the argument that literate women of the Renaissance sought to oppose social constraints imposed on them

C. show a way in which Krontiris's work illustrates a "cautiously optimistic" assessment of Renaissance women's achievements

D. summarize Krontiris's view of the effect of literacy on the lives of upper- and middle-class Renaissance women

E. illustrate the way in which Krontiris's study differs from the studies done by Burckhardt and Kelly

605. The author of the passage implies that the women studied by Krontiris are unusual in which of the following ways?

A. They faced obstacles less formidable than those faced by other Renaissance women

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A. They faced obstacles less formidable than those faced by other Renaissance women.

B. They have been seen by historians as more interesting than other Renaissance women.

C. They were more concerned about recording history accurately than were other Renaissance women.

D. Their perceptions are more likely to be accessible to historians than are those of most other Renaissance women.

E. Their concerns are likely to be of greater interest to feminist scholars than are the ideas of most other Renaissance women.

Line When asteroids collide, some collisions cause an asteroid to spin faster; others slow it down. If asteroids are all monoliths—single rocks—undergoing random collisions, a graph of their rotation rates (5) should show a bell-shaped distribution with statistical "tails" of very fast and very slow rotators. If asteroids are rubble piles, however, the tail representing the very fast rotators would be missing, because any loose aggregate spinning faster than once every few (10) hours (depending on the asteroid's bulk density) would fly apart. Researchers have discovered that all but five observed asteroids obey a strict limit on rate of rotation. The exceptions are all smaller than 200 meters in diameter, with an abrupt cutoff for (15) asteroids larger than that.

The evident conclusion—that asteroids larger than 200 meters across are multicomponent structures or rubble piles—agrees with recent computer modeling of collisions, which also finds a transition at that (20) diameter. A collision can blast a large asteroid to bits, but after the collision, those bits will usually move slower than their mutual escape velocity. Over several hours, gravity will reassemble all but the fastest pieces into a rubble pile. Because collisions among (25) asteroids are relatively frequent, most large bodies have already suffered this fate. Conversely, most small asteroids should be monolithic, because impact fragments easily escape their feeble gravity.

**Questions 606–609 refer to the passage.**

606. The passage implies which of the following about the five asteroids mentioned in line 12?

A. Their rotation rates are approximately the same.

B. They have undergone approximately the same number of collisions.

C. They are monoliths.

D. They are composed of fragments that have escaped the gravity of larger asteroids.

E. They were detected only recently.

607. The discovery of which of the following would call into question the conclusion mentioned in line 16?

A. An asteroid 100 meters in diameter rotating at a rate of once per week

B. An asteroid 150 meters in diameter rotating at a rate of 20 times per hour

C. An asteroid 250 meters in diameter rotating at a rate of once per week

D. An asteroid 500 meters in diameter rotating at a rate of once per hour

E. An asteroid 1,000 meters in diameter rotating at a rate of once every 24 hours

608. According to the passage, which of the following is a prediction that is based on the strength of the gravitational attraction of small asteroids?

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attraction of small asteroids?

A. Small asteroids will be few in number.

B. Small asteroids will be monoliths.

C. Small asteroids will collide with other asteroids very rarely.

D. Most small asteroids will have very fast rotation rates.

E. Almost no small asteroids will have very slow rotation rates.

609.

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